

Kirubaraj Manimathavan

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EXPERIENCE

Amazon

January 2025 – Present

Software Development Engineer II – Applied AI

- Early engineer on the **Agentic AI Platform** powering 1M+ AI agents/day across Amazon, saving **\$950M+** annually.
- Architected a framework-agnostic **Agentic Tool Integration layer** enabling direct API invocation as a lightweight alternative to MCP, cutting tool onboarding from **2 weeks to 10 minutes** and boosting tool coverage by **44%**.
- **Led** design, development of extensible **Event-based Triggers SDK**, replacing manual workflows with self-service automation — powering **30%** of all platform executions at **25k+** events/second, later extended for tool-level triggering.
- Built a **Persistent Agent Memory** layer using open-source **mem0** with tenant-isolated vector stores, enabling agents to retain and retrieve context across executions, improving agent success rate by over **25%**.

Troutwood

January 2024 – December 2024

AI Software Engineer – Contract

- **Led** a 4-member team to build **Langchain LLM Agents** which reduced research time for Financial Advisors by **50%**.
- Designed and implemented an AI Agent leveraging open source multimodal LLMs, embedding models, and vector databases for **RAG**. Deployed system on AWS EKS to deliver enterprise-level **security** and **privacy** for clients.
- Developed **API** based Tools and **Parallel Tool Calling** for Langchain Bedrock increasing chat **performance** by **8X**.

Oracle

November 2022 – July 2023

Senior Software Engineer

- **Led** a cross-functional team of 3 in **technical design, development, and deployment** (CI/CD Pipeline) of multiple full-stack enterprise software projects for Oracle ERP, resulting in successful onboarding of **3** enterprise customers.
- Spearheaded architecture design and end-to-end development of search and auto-fill features using **Elasticsearch** with real-time data ingestion and monitoring with **Logstash** and **Kibana**, boosting search result click rate by **75%**.
- Transformed Monolith/JSF architecture of Flexfield component to independently deployable **ReactJS** component with backend **NodeJS microservices** deployed on Kubernetes, enabling portability and **scalability** across **12** teams.

Software Engineer

August 2020 – November 2022

- Unified multiple custom **PL/SQL** Packages and Data Models into seeded parametrized offerings avoiding code redundancy and increasing **maintainability**, resulting in decrease of inbound bugs by **27%** in the reporting product.
- Reconstructed the report visualization engine by parallelizing **SOA** service calls, redesigning ReactJS renderer and tuning the underlying **SQL queries** by optimizing their joins, indexes and query hints reducing rendering time by **70%**.
- Resolved **77** customer issues, enhancing **performance** and **availability** of business-critical reports, with **30%** Sev-1 escalations. Worked and communicated with customers during on-call rotation with average turnaround of **8 hours**.

Eat From Home (TOVO Technologies Private Limited) [\[GitHub\]](#)

August 2018 – August 2020

Co-Founder and CTO

- Designed and developed an **Android** Mobile app for on-demand e-commerce with **45,000+** installs and **3,000+** DAUs.
- Built and managed a **NodeJS REST API** server for integration of user, rider and chef apps using Shopify **GraphQL** APIs, Razorpay **webhooks**, **Firebase** database for efficient order management ensuring real-time order tracking.
- Optimized order allocation system through order batching using **Kafka** and **Samza** reducing delivery cost by **35%**.
- **Led** and **mentored** 5 engineers/designers in development of rider/chef apps and ensured on-time delivery.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Software Engineering

December 2024

Coursework - Deep Learning, Cloud Computing, Distributed Systems, Data Structures, Algorithms, Software Architecture

SKILLS

AI/ML – Python | Agent Orchestration, Prompt Engineering, RAG, Embeddings, Transformers, Fine-tuning, AI Evaluation
Platform/Infra – PyTorch, LangChain, Bedrock AgentCore, Agent Memory | AWS, CloudFormation, Kubernetes, CI/CD
Backend/Data – Java, Go, NodeJS, MCP, Tool Use | DynamoDB, MongoDB, Vector DB, Redis, Elasticsearch, Kafka
Full Stack – Typescript, ReactJS, NextJS | REST API, Web Sockets, SSE, gRPC, GraphQL | HTML5, CSS3

PROJECTS

Integrating Short-Term News Sentiment To Stock Prediction [\[GitHub\]](#), [Thesis](#), [Demo](#)

January 2024 – May 2024

- A short-term stock prediction model using **attention techniques** to integrate news sentiment to historical stock data.
- Implemented **3** variations of **Transformer** Architecture on baseline **CNN-LSTM** model, improving MSE loss by **94.8%**.

Social Network Page Ranking using Iterative Processing on Spark [\[GitHub\]](#)

September 2023 – October 2023

- Developed Page Rank algorithm in **Scala**, with **RDD** caching and optimized network shuffles reducing runtime by **67%**.